Answers to Exercises 1

1.

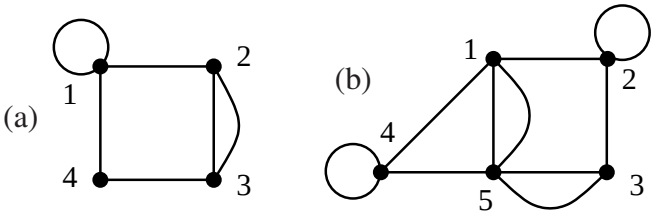
$$\text{(a)}\begin{pmatrix} 0 & 1 & 1 & 1 \\ 1 & 2 & 2 & 1 \\ 1 & 2 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{pmatrix} \quad \text{(b)}\begin{pmatrix} 0 & 1 & 0 & 0 & 1 \\ 1 & 2 & 2 & 0 & 0 \\ 0 & 2 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 1 & 2 \end{pmatrix}$$

Adjacency lists:

(a)
1 → 2 → 3 → 4
2 → 1 → 2 → 3 → 3 → 4
3 → 1 → 2 → 2 → 4
4 → 1 → 2 → 3

(b)
1 → 2 → 5
2 → 1 → 2 → 3 → 3
3 → 2 → 2 → 4 → 5
4 → 3 → 5
5 → 1 → 3 → 4 → 5

2.



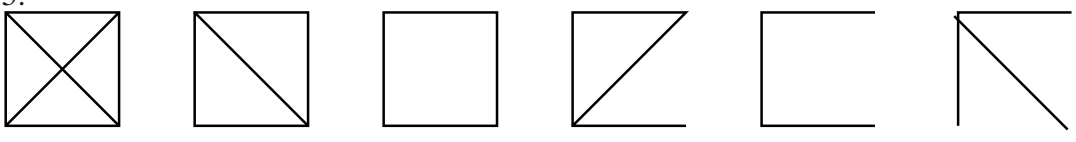
3. The righthand graph has a node of degree two which is adjacent to two nodes of degree three. There is no such node in the lefthand graph.

4. (a),(b) Node 1 may be mapped to any of 7,8,10,11 (all nodes of degree 3). The rest of the isomorphism is then determined. The four possible isomorphisms are:

1	7	8	10	11
2	12	9	9	12
3	11	10	8	7
4	8	7	11	10
5	9	12	12	9
6	10	11	7	8

(c) Four.
In general, if G_1 is isomorphic to G_2 then the number of automorphisms of G_1 is the same as the number of isomorphisms between G_1 and G_2.

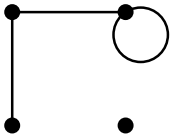
5.



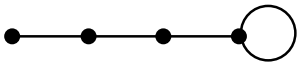
Graph 1 is K_4, the complete graph with four nodes. Graph 2 is called the 'diamond' graph. Graph 3 is C_4, the cycle graph with four nodes, also denoted $K_{2,2}$ when seen as a bipartite graph. Graph

4 is called the 'paw' graph. Graph 5 is P_4, the path graph with four nodes. Graph 6 is S_4, the star graph with four nodes, also denoted $K_{1,3}$ when seen as a bipartite graph.

6.
(a)



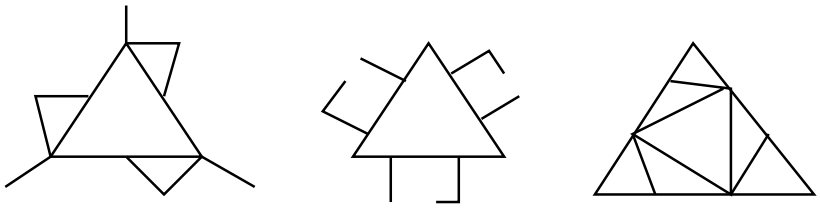
Note that all the nodes have different degrees, so that any automorphism must be the identity, since isomorphism preserves degrees of nodes.
(b)



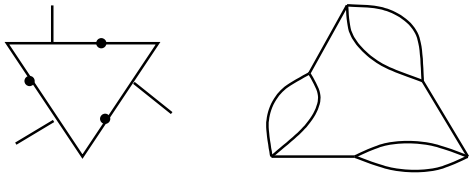
This is, of course, a valid answer for (a).
Here there are two nodes with degree 2, but if we consider the degrees of their neighbours we see that an automorphism cannot swap these two nodes.
(c) There is no simple graph (connected or otherwise) with four nodes and no automorphism which is not the identity. So we need to have loops to get a solution.
Justification. First consider disconnected simple graphs. We can have at most one isolated node, or else we could swap isolated nodes to get a non-identity automorphism. With one isolated node the remaining three nodes form either a triangle K_3 or a path graph P_3. Non-identity automorphisms are possible in both cases.
The only other possibility is two disconnected components each with two nodes and one arc ($P_2 + P_2$). Non-identity automorphisms are possible in this case.
Now consider connected simple graphs. The only possibilities are the six in Question 5. This can be seen by starting with graph 1 (K_4) and removing one arc (graph 2), and then a further arc (graphs 3 and 4), and then a further arc (graphs 5 and 6). At least three arcs are needed to connect the graph. Non-identity automorphisms are possible in all six cases.

7. Recommended method:
We consider how many ways each node can be assigned:
Node 2 must be fixed (only node of degree 6).
Node 1 can go to 6 positions. Fix 1. Then 7 is fixed.
Node 3 can go to 4 positions. Fix 3. Then 4 is fixed.
Node 5 can go to 2 positions. Fix 5. Then 6 is fixed.
This gives $1.6.1.4.1.2.1 = 48$ automorphisms in all.
Alternatively:
Node 2 must be fixed. There are $3! = 6$ permutations of the triangles. Then there are 3 independent flips of each triangle, giving 2^3 possibilities. Answer $6.8 = 48$.

8. A triangle seems like the obvious starting point, but a triangle has 6 automorphisms, since it has reflective symmetry as well as rotational symmetry. So we remove the reflective symmetry while preserving the rotational symmetry. Here are three simple graphs (i.e. no loops or parallel arcs):



It is not possible to use fewer than nine nodes, and the third graph above has the fewest arcs among nine-node examples.
The following don't work (they have 6 automorphisms and $6.2.2.2 = 48$ automorphisms, with swapping of parallel arcs):



9. Graphs which are isomorphic to their complements are said to be *self-complementary*.
(a) G_4 is the path graph P_4.
G_5 can be the cycle graph C_5, or a triangle with arcs dangling from two nodes.
(b) Suppose G has six nodes and $\overline{G}$ is isomorphic to G. Then G and $\overline{G}$ have the same number of arcs. The total number of arcs in both must be even. Also since $\overline{G}$ is simple, G is simple. So the total number of arcs must be the same as for K_6, namely $6.5/2 = 15$. Since this is odd, we have a contradiction.
[We can generalise this argument to show that if G is a self-complementary graph with n nodes, then it must have $n(n-1)/4$ arcs. Hence either n or $n-1$ is divisible by 4, that is, n must be congruent to 0 or 1 mod 4.]

10. (a) 16.
(b) 3 and 4 are fixed since they are the only nodes with degrees 3, 5 respectively.
1 has 2 possible positions - fix 1.
2 has 1
5 has 4 - fix 5
6 has 1
7 has 2 - fix 7
8 has 1
Product 2.4.2 = 16.